Characterization of frequency-to-place mismatch using frequency shifted music: a vocoder study

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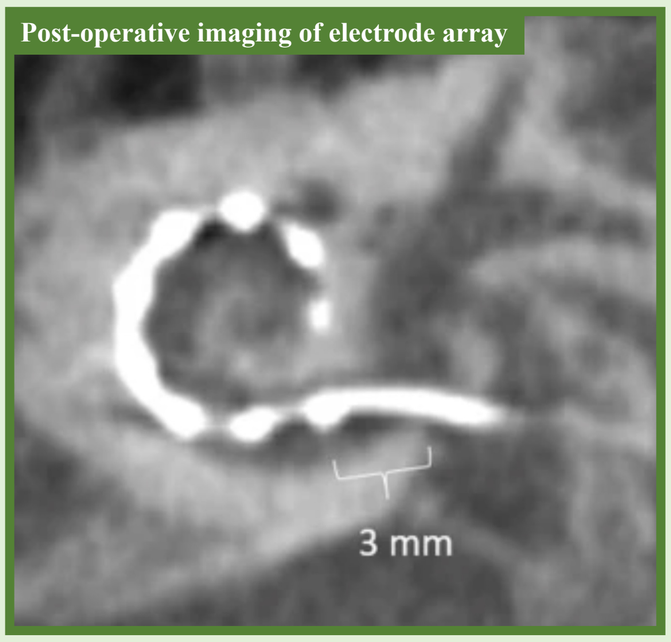
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Introduction

In more recent years a greater emphasis has been placed on improving **the sound quality** for individuals using cochlear implants (CIs) . Indeed, many post-lingually deafened adults with CIs have speech recognition performance close to normal hearing (NH) listeners in quiet environments, but yet complain about poor sound quality, especially for music.

The contribution of CI technology to the degradation of sound quality has been well documented, but little is known about the relative importance of different aspects relating to the technology and how they interact with the personal preferences and hearing characteristics of the individual. One individual factor that is suspected to contribute largely to the sound quality degradation is the **frequency-to-place mismatch (FTPM)** induced by the incomplete insertion of the electrode array inside the cochlea. It is defined as a deviation between the characteristic frequency and the allocation frequency assigned during CI fitting.

Post-operative imaging of electrode array



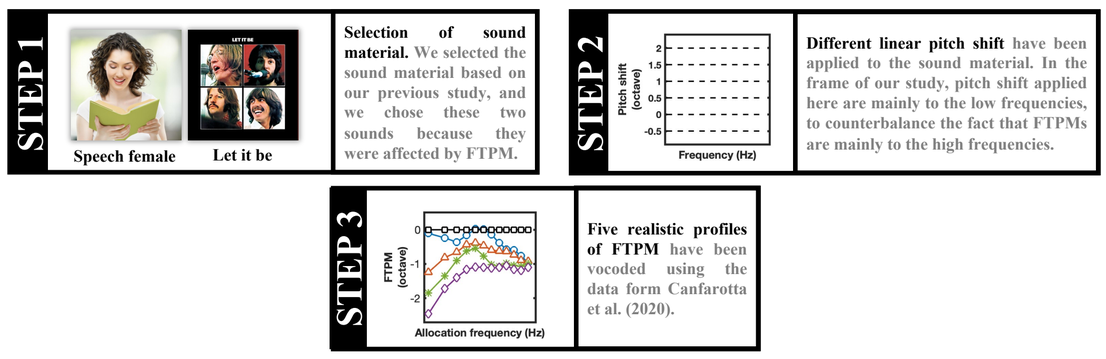
Whereas much of the speech perception improvement occurs during the first months after CI activation, recent studies have shown that a significant FTPM could negatively affect this adaptation process. Alongside the improvements in speech perception over time from device activation patients often also change their reports of sound quality, from “high pitched”, “screechy”, “dalek-like” to more “normal” or “natural” after extensive listening experience.

The only reliable way to measure the accuracy of the positioning of the electrode array inside the cochlea is through imaging, which involves comparing the position of simulated neurons to the actual location of the electrodes. However, this type of evaluation is not typically utilized during routine medical checkups.

This research investigates how pre-modifications of speech and music material can be used to explore the FTPM.

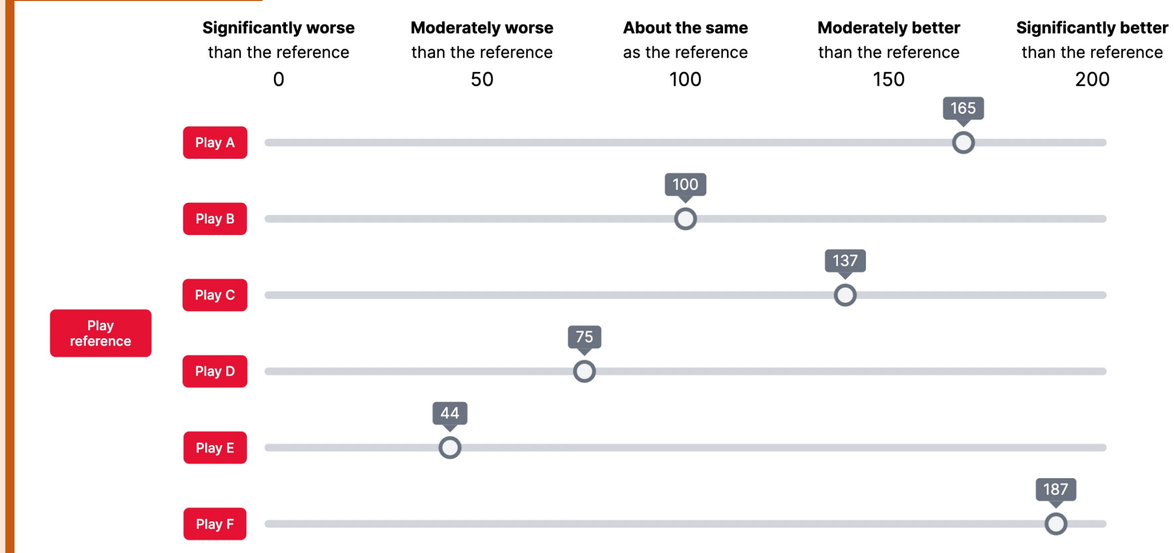
Material & Method

Sound processing

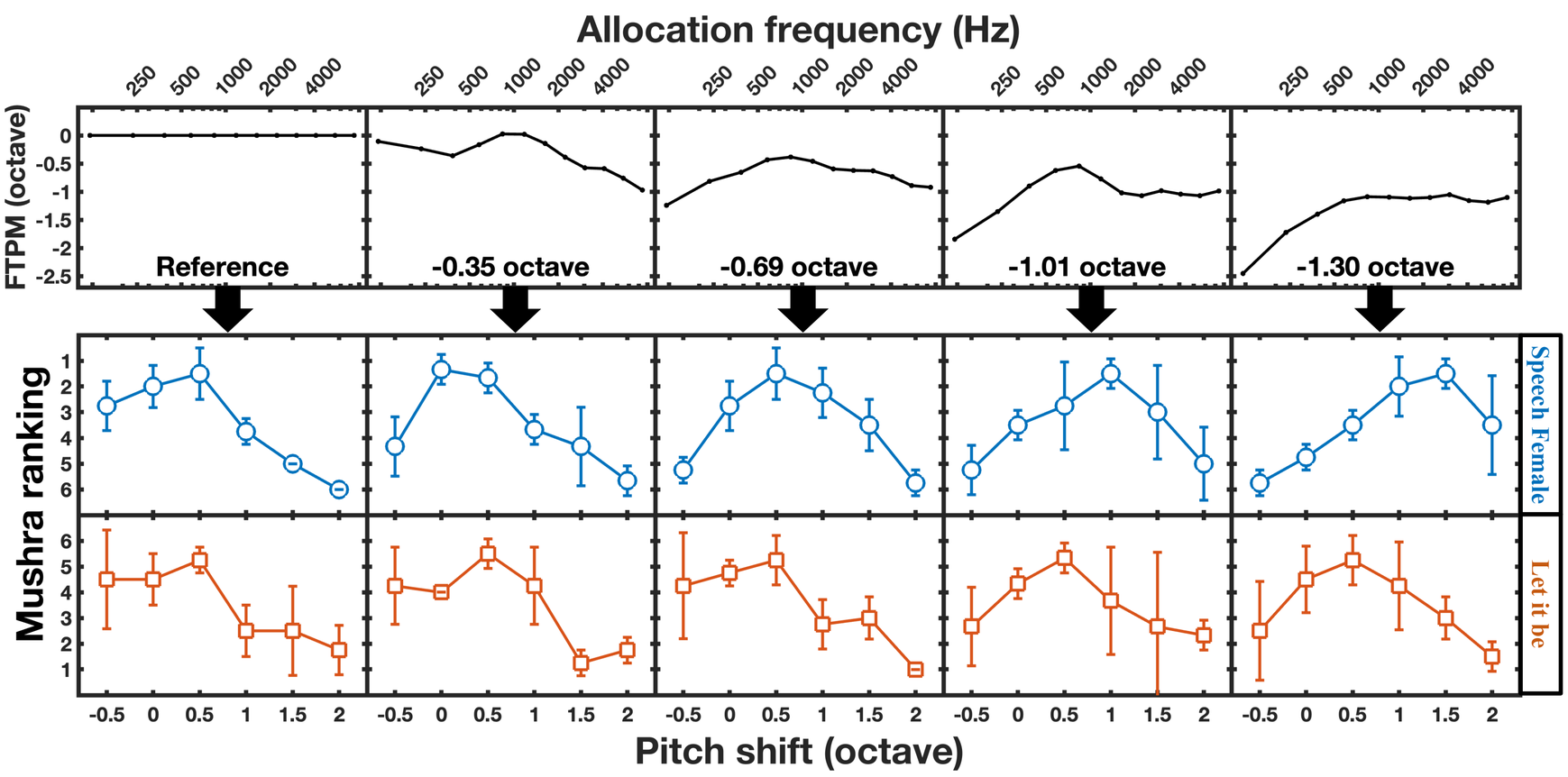


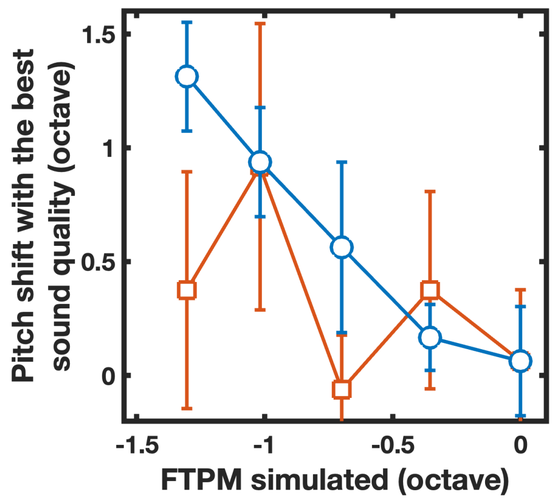
- Four normal hearing listeners tested
- Two sound materials tested
- Six pitch levels applied
- Five degrees of FTPM simulated
- One noise vocoder tested

MUSHRA method



Results





Results at a glance

1. Based on our preliminary data, the speech used appears to be an effective predictor of the degree of FTPM simulated.
2. At this stage, there is no clear trend regarding the effect of the music "Let it be"

Next steps

1. Increase the sample size by testing more participants.
2. Expand the range of music tested.
3. Improve the analysis methodology.
4. Modify the MUSHRA to incorporate an adaptive approach that is appropriate for CI users